

Stat 184 Homework 4.3

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1 Busiest Airports

1.1 Busiest Airports Table

Table 1: Busiest Airports by Annual Passenger Traffic (Millions), 2020-2023

Airport	Code	2023	2022	2021	2020
Hartsfield–Jackson Atlanta International Airport	ATL	106.30	108.07	104.65	93.70
Dubai International Airport	DXB	95.20	92.33	86.99	66.07
Dallas Fort Worth International Airport	DFW	85.66	87.82	81.76	73.36
Heathrow Airport	LHR	84.46	83.88	79.18	61.61
Frankfurt Airport	FRA	63.19	61.56	59.36	48.92
Beijing Daxing International Airport	PKX	53.62	49.44	NA	NA

1.2 Busiest Airports Plot

1.3 Buisiest Airports Narrative

As shown in Table 1, the table gives a visual representation of airport travel through six selected airports for the years 2020-2025. The heatmap uses both colors and numbers to highlight changes not only in a single airport’s travelers, but also to compare the differences across the six different airports throughout the selected years.

The Figure 1 visualizes the amount of airport travelers through the six selected airports using a scatter and line plot. This graph allows readers to better compare the raw amount of passengers between the airports and how the amount of travelers varied year-by-year.

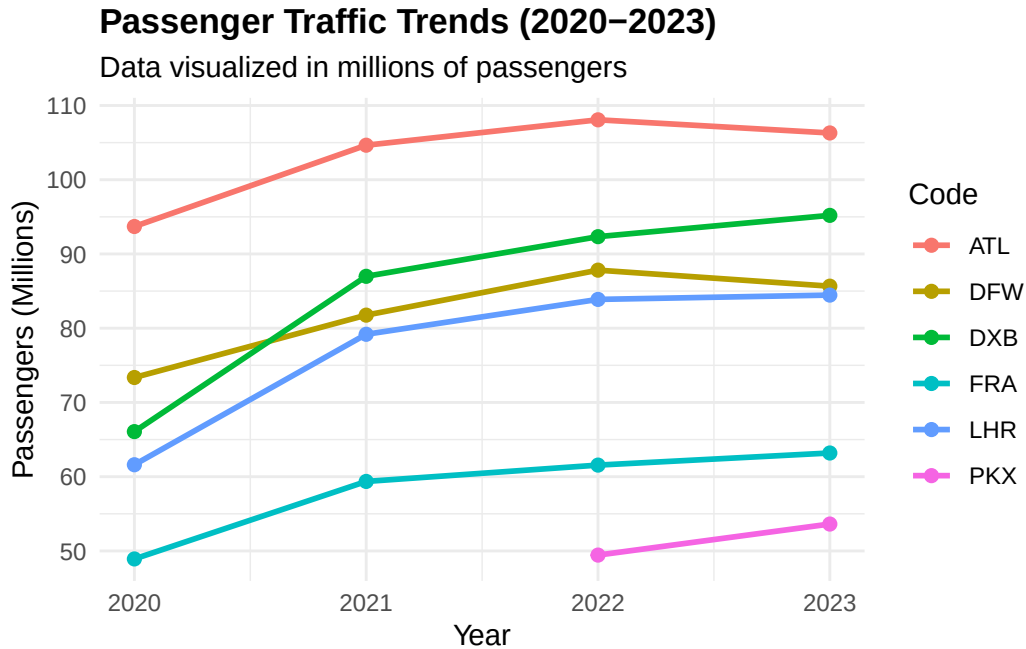


Figure 1: Passenger Traffic Trends (2020-2023) for Selected Major Global Hubs.

2 Monte Carlo Number Integration

2.1 Monte Carlo Number Integration Plot

2.2 Monte Carlo Number Integration Narrative

The Figure 2 visualizes how number integration works using simulated points. It can be seen that randomized points get plotted below or above the line $f(x) = x^2$, and can be distinguished by their different colors. The more amount of points that are plotted, the clearer the difference between above and below the line $f(x) = x^2$ becomes.

3 Planning and Prompting GenAI

3.1 Planning How to Wrangle the Data

Goal: The goal of this section is to first tidy the calcium.csv data file, then create an appropriate plot/graph that will allow readers to evaluate the difference in ulnar calcium measurements across time and the two treatment groups.

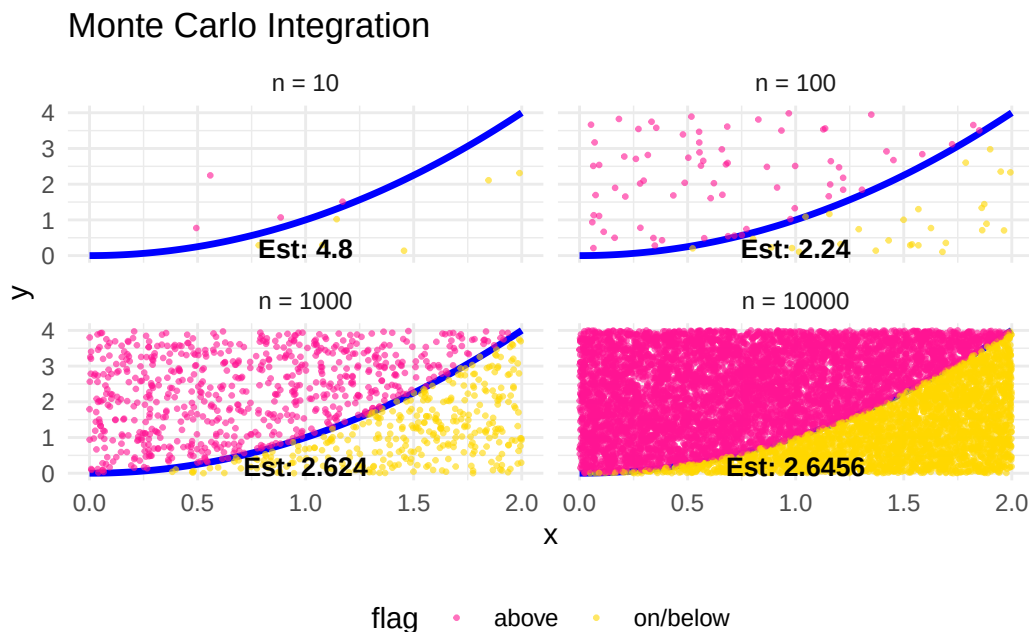


Figure 2: Monte Carlo Integration of $f(x) = x^2$ showing the convergence of area estimates as n increases.

Nouns: calcium.csv, GenAI(Antropic Claude), Rstudio, Quarto Markdown Document

Verbs: tidy, prompt, code, generate, render

Steps: 1. Remove the first row of data, which are the title's of each column. 2. Move columns so every row becomes a case (a single woman). Move columns 5-8(diet/exercise group) under the first four(null treatment group). 3. Add a column named "Treatment Group" identifying the treatment group of each woman. The first 15 rows should be labeled as null treatment, and the next 16 should be labeled as the diet/exercise group. 4. Rename columns to appropriate names(column 1 = initial, column 2 = Year 1, column 3 = Year 2, column 4 = Year 3) 5. Create an appropriate plot/graph of the data using the kable package.

This was the output generated by Claude:

Table 2: Tidy Calcium Dataset

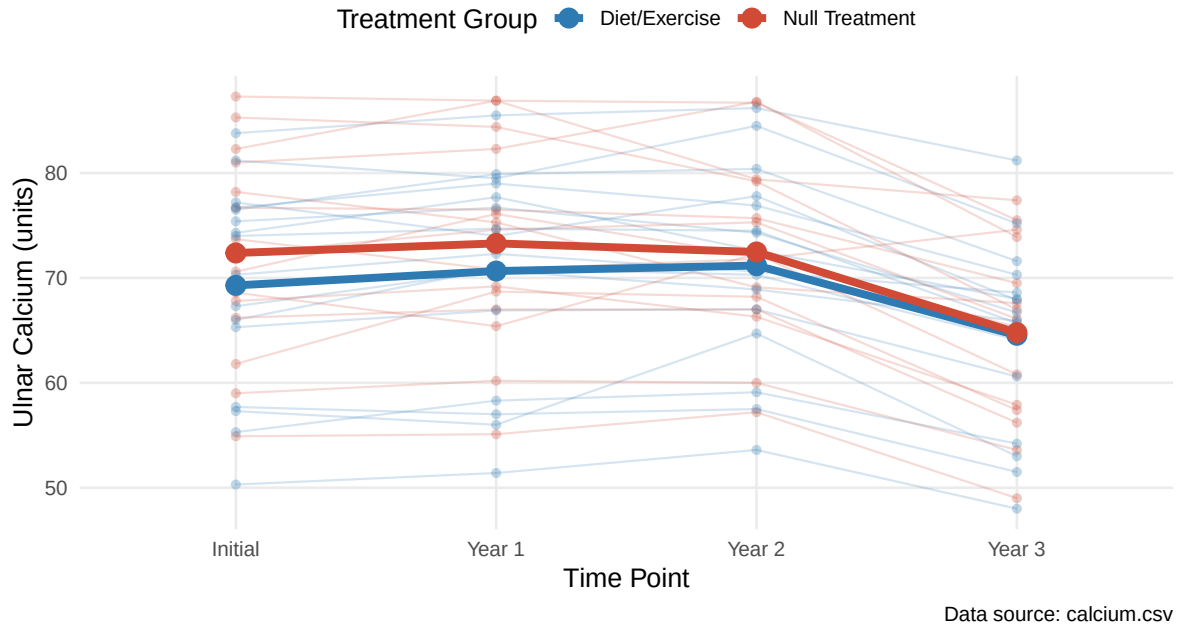
Initial	Year 1	Year 2	Year 3	Treatment Group
Null Treatment				
87.3	86.9	86.7	75.5	Null Treatment
59.0	60.2	60.0	53.6	Null Treatment
76.7	76.5	75.7	69.5	Null Treatment

70.6	76.1	72.1	65.3	Null Treatment
54.9	55.1	57.2	49.0	Null Treatment
78.2	75.3	69.1	67.6	Null Treatment
73.7	70.8	71.8	74.6	Null Treatment
61.8	68.7	68.2	57.4	Null Treatment
85.3	84.4	79.2	67.0	Null Treatment
82.3	86.9	79.4	77.4	Null Treatment
68.6	65.4	72.3	60.8	Null Treatment
67.8	69.2	66.3	57.9	Null Treatment
66.2	67.0	67.0	56.2	Null Treatment
81.0	82.3	86.8	73.9	Null Treatment
72.3	74.6	75.3	66.1	Null Treatment
Diet/Exercise				
83.8	85.5	86.2	81.2	Diet/Exercise
65.3	66.9	67.0	60.6	Diet/Exercise
81.2	79.5	84.5	75.2	Diet/Exercise
75.4	76.7	74.3	66.7	Diet/Exercise
55.3	58.3	59.1	54.2	Diet/Exercise
70.3	72.3	70.6	68.6	Diet/Exercise
76.5	79.9	80.4	71.6	Diet/Exercise
66.0	70.9	70.3	64.1	Diet/Exercise
76.7	79.0	76.9	70.3	Diet/Exercise
77.2	74.0	77.8	67.9	Diet/Exercise
67.3	70.7	68.9	65.9	Diet/Exercise
50.3	51.4	53.6	48.0	Diet/Exercise
57.7	57.0	57.5	51.5	Diet/Exercise
74.3	77.7	72.6	68.0	Diet/Exercise
74.0	74.7	74.5	65.7	Diet/Exercise
57.3	56.0	64.7	53.0	Diet/Exercise

4 Step 5: Visualize Ulnar Calcium Over Time by Treatment Group

Ulnar Calcium Measurements Over Time

Individual trajectories (faint) with group means (bold) by treatment group



4.1 Interpretation

The plot above displays each woman's ulnar calcium trajectory from **Initial** through **Year 3**, with bold lines representing group means. This allows readers to evaluate:

- **Within-group trends** over time (do calcium levels decline across years?)
- **Between-group differences** (does the Diet/Exercise group maintain or improve calcium levels relative to the Null Treatment group?)

5 Using a Generic Prompt with GenAI

These are the responses from Claude after using a generic prompt:

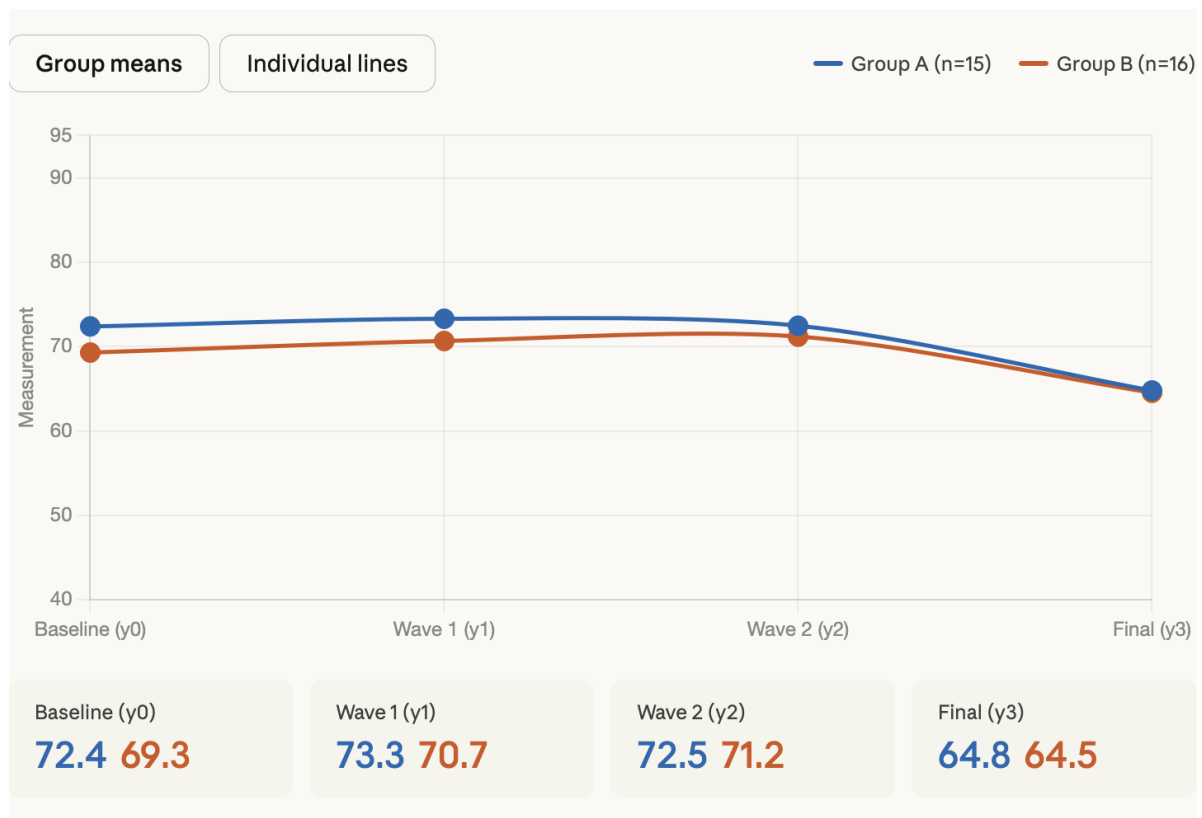


Figure 3: Claude Plot of Group Mean Calcium Data Using a Generic Prompt

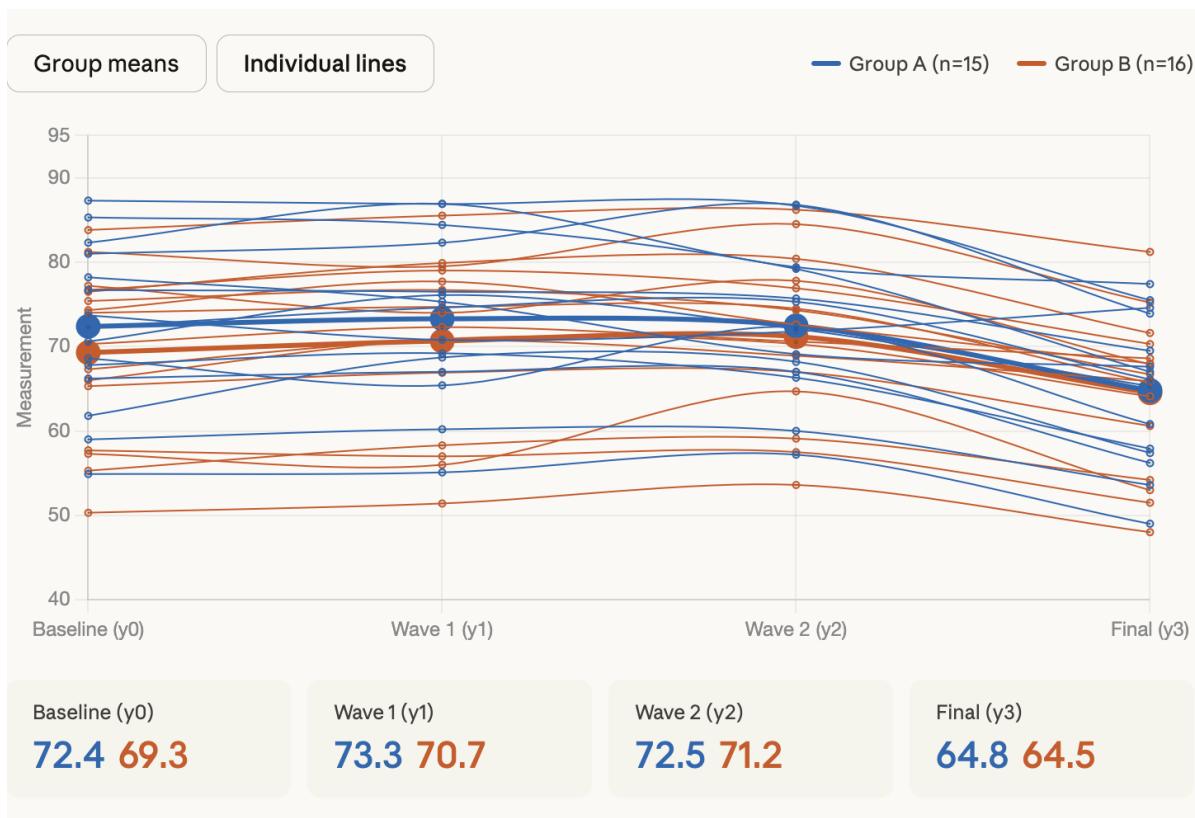


Figure 4: Claude Plot of Individual Mean Calcium Data Using a Generic Prompt

Claude's Interpretation of the Data:

Here's what stands out in the data: Both groups follow the same pattern — values stay relatively stable across the first three waves ($y_0 \rightarrow y_1 \rightarrow y_2$), then drop noticeably at the final measurement (y_3). This sharp decline at y_3 is the dominant signal. Group A starts slightly higher (baseline mean ~ 72.4 vs. ~ 69.3 for Group B), but both groups converge to nearly identical final values (~ 64.8 and ~ 64.5). That convergence is interesting — Group A drops about 7.5 points while Group B drops about 4.8. There's substantial individual variation — hit the "Individual lines" toggle to see this. Some participants stay high throughout, others are consistently low, and a few show unusual trajectories (one in Group A actually rises at y_3 while most fall). If you can share what the groups represent (e.g., treatment vs. control, or different diets/supplements) and what's being measured, I can dig into the statistics more precisely — things like whether the group difference at y_3 is significant, or which individuals are outliers. Version: Claude Sonnet 4.6 Extended (This chat took place on Tuesday, March 31st, 2026.)

5.1 GenAI Narrative

There is a difference between the results, although the response from the generic prompt could be considered a good start. Using a more detailed prompt allowed the prompter to ensure that Claude would follow a specific path when creating its output. This can be important especially if the prompter has methods that they know they want to use/follow, as oppose to allowing GenAI to have free reign. Having a more specific prompt also ensures that the output is more specific and tailored to what the prompter requested. Claude's interpretation of the data is also very broad/basic, listing obvious facts from the data and gleaning no deeper insights. This may be because GenAI is not as adept at understanding broader context and/or analyzing images, something that a person would be better at doing.

6 GenAI Usage Appendix

1. In this document I used GenAI tools (Claude specifically) for the GenAI section, as well as writing various alt-texts for the plots/graphs. I also consulted Claude in various sections with miscellaneous items, smaller prompts such as "how to add captions to this R studio plot" and "what is this error message and why isn't my qmd file creating a pdf".
2. For the GenAI sections I used Claude's Sonnet 4.6 on 4/10/2026. I used additional help with smaller issues such as errors and formatting the qmd document from Claude on 4/12/2026 and 4/13/2026.

7 Code Appendix

```
library(tidyverse)
library(rvest)
library(knitr)
library(kableExtra)

wiki_url <- "https://en.wikipedia.org/wiki/List_of_busiest_airports_by_passenger_traffic"

wikiRawList <- read_html(x = wiki_url) |>
  html_elements(css = "table.wikitable") |>
  html_table()

airport_codes <- c("ATL", "FRA", "PKX", "DXB", "DFW", "LHR")
code_pattern <- paste(airport_codes, collapse = "|")
```



```

data_2023_raw <- wikiRawList[[1]]
colnames(data_2023_raw) <- make.unique(colnames(data_2023_raw))
data_2023 <- data_2023_raw |>
  rename(Code = contains("Code") | contains("IATA")) |>
  filter(str_detect(Code, code_pattern)) |>
  select(Airport, Code, Passengers = contains("2023") | contains("Passenger")) |>
  mutate(Year = 2023)

data_2022_raw <- wikiRawList[[2]]
colnames(data_2022_raw) <- make.unique(colnames(data_2022_raw))
data_2022 <- data_2022_raw |>
  rename(Code = contains("Code") | contains("IATA")) |>
  filter(str_detect(Code, code_pattern)) |>
  select(Airport, Code, Passengers = contains("2022") | contains("Passenger")) |>
  mutate(Year = 2022)

data_2021_raw <- wikiRawList[[3]]
colnames(data_2021_raw) <- make.unique(colnames(data_2021_raw))
data_2021 <- data_2021_raw |>
  rename(Code = contains("Code") | contains("IATA")) |>
  filter(str_detect(Code, code_pattern)) |>
  select(Airport, Code, Passengers = contains("2021") | contains("Passenger")) |>
  mutate(Year = 2021)

data_2020_raw <- wikiRawList[[4]]
colnames(data_2020_raw) <- make.unique(colnames(data_2020_raw))
data_2020 <- data_2020_raw |>
  rename(Code = contains("Code") | contains("IATA")) |>
  filter(str_detect(Code, code_pattern)) |>
  select(Airport, Code, Passengers = contains("2020") | contains("Passenger")) |>
  mutate(Year = 2020)

busiestAirportsTidy <- bind_rows(data_2023, data_2022, data_2021, data_2020) |>
  mutate(
    Passengers_Millions = round(parse_number(as.character(Passengers)) / 1e6, 2),
    Airport = str_remove_all(Airport, "\\.[*\\d+\\]"),
    Airport = str_squish(Airport),
    Code = str_extract(Code, code_pattern)
  ) |>
  select(Airport, Code, Year, `Passengers (Millions)` = Passengers_Millions)

busiestAirportsTidy |>

```

```

pivot_wider(names_from = Year, values_from = `Passengers (Millions)`) |>
kable(booktabs = TRUE, align = "lcccc") |>
kable_styling(latex_options = c("striped", "hold_position"), full_width = FALSE)
ggplot(busiestAirportsTidy, aes(x = Year, y = `Passengers (Millions)`, color = Code, group =
geom_line(linewidth = 1) +
geom_point(size = 2) +
scale_x_continuous(breaks = 2020:2023) +
theme_minimal() +
labs(title = "Passenger Traffic Trends (2020-2023)",
      subtitle = "Data visualized in millions of passengers",
      y = "Passengers (Millions)",
      x = "Year") +
  theme(plot.title = element_text(face = "bold")))
library(tidyverse)
library(patchwork)

f <- function(x) x^2
x_min <- 0
x_max <- 2
y_min <- 0
y_max <- 4
rect_area <- (x_max - x_min) * (y_max - y_min)

n_values <- c(10, 100, 1000, 10000)

all_sim_data <- map_df(n_values, function(n_val) {
  data.frame(
    x = runif(n_val, x_min, x_max),
    y = runif(n_val, y_min, y_max),
    n = n_val
  )
}) |>
mutate(
  flag = if_else(y <= f(x), "on/below", "above"),
  facet_label = factor(paste0("n = ", n), levels = paste0("n = ", n_values))
)

estimates <- all_sim_data |>
group_by(facet_label) |>
summarize(
  p = mean(flag == "on/below"),
  est = p * rect_area

```

```

)

ggplot(all_sim_data) +
  stat_function(fun = f, color = "blue", linewidth = 1.2, xlim = c(x_min, x_max)) +
  geom_point(aes(x, y, color = flag), size = 0.5, alpha = 0.6) +
  geom_text(data = estimates,
            aes(x = (x_min + x_max)/2, y = 0.2, label = paste("Est:", round(est, 4))),
            size = 3.5, fontface = "bold") +
  facet_wrap(~facet_label, nrow = 2) +
  scale_color_manual(values = c("above" = "deeppink", "on/below" = "gold")) +
  theme_minimal() +
  labs(title = "Monte Carlo Integration",
       x = "x", y = "y") +
  theme(legend.position = "bottom")
library(tidyverse)
library(knitr)
library(kableExtra)

# Read the CSV, skipping the first row (original column headers)
calcium_raw <- read.csv("~/Downloads/calcium.csv", header = FALSE, skip = 1)

# Split into the two groups (columns 1-4 = null treatment, columns 5-8 = diet/exercise)
null_group <- calcium_raw[1:15, 1:4]
diet_group <- calcium_raw[1:16, 5:8]

# Add Treatment Group label
null_group$`Treatment Group` <- "Null Treatment"
diet_group$`Treatment Group` <- "Diet/Exercise"

# Rename columns to appropriate time-point names
col_names <- c("Initial", "Year 1", "Year 2", "Year 3", "Treatment Group")
colnames(null_group) <- col_names
colnames(diet_group) <- col_names

# Stack the two groups (every row = one woman)
calcium_tidy <- bind_rows(null_group, diet_group)

# Preview the tidy data
kable(calcium_tidy, caption = "Tidy Calcium Dataset") |>
  kable_styling(
    bootstrap_options = c("striped", "hover", "condensed", "bordered"),
    full_width = FALSE,

```

```

    position = "center"
  ) |>
  row_spec(0, bold = TRUE, background = "#4A7C9E", color = "white") |>
  pack_rows("Null Treatment", 1, 15) |>
  pack_rows("Diet/Exercise", 16, 31)
# Pivot to long format for plotting
calcium_long <- calcium_tidy |>
  mutate(ID = row_number()) |>
  pivot_longer(
    cols = c("Initial", "Year 1", "Year 2", "Year 3"),
    names_to = "Time",
    values_to = "Calcium"
  ) |>
  mutate(Time = factor(Time, levels = c("Initial", "Year 1", "Year 2", "Year 3")))

# Compute group means for overlay
calcium_means <- calcium_long |>
  group_by(`Treatment Group`, Time) |>
  summarise(Mean_Calcium = mean(Calcium, na.rm = TRUE), .groups = "drop")

# Plot
ggplot(calcium_long, aes(x = Time, y = Calcium, color = `Treatment Group`, group = interaction(
  # Individual trajectories (faint)
  geom_line(alpha = 0.2, linewidth = 0.5) +
  geom_point(alpha = 0.3, size = 1.5) +
  # Group mean trajectories (bold)
  geom_line(
    data = calcium_means,
    aes(x = Time, y = Mean_Calcium, group = `Treatment Group`, color = `Treatment Group`),
    linewidth = 1.8, inherit.aes = FALSE
  ) +
  geom_point(
    data = calcium_means,
    aes(x = Time, y = Mean_Calcium, color = `Treatment Group`),
    size = 4, inherit.aes = FALSE
  ) +
  scale_color_manual(values = c("Null Treatment" = "#D04A35", "Diet/Exercise" = "#2E7BB4"))
  labs(
    title = "Ulnar Calcium Measurements Over Time",
    subtitle = "Individual trajectories (faint) with group means (bold) by treatment group",
    x = "Time Point",
    y = "Ulnar Calcium (units)",

```

```
    color = "Treatment Group",
    caption = "Data source: calcium.csv"
) +
theme_minimal(base_size = 13) +
theme(
  plot.title = element_text(face = "bold", size = 15),
  plot.subtitle = element_text(color = "gray40", size = 11),
  legend.position = "top",
  panel.grid.minor = element_blank()
)
```